

Hedging and Options

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Set Up

Suppose that we have a European call option of price C_t on an underlying stock with price S_t that does not pay dividends. Assume the initial price of the equity is $S_0 = 90$, the strike price is $K = 100$, the risk-free rate $r = 0$, and the volatility of the underlying stock is $\sigma = 20\%$. Say the call expires in one-year, i.e. $T = 1$. Under the Black-Scholes framework, the initial price of the call is $C_0 \approx 3.59$.

Taylor's Theorem

Consider the price of the call at time t as a function of S_t , i.e. let

$$f_t(S_t) = C_t.$$

We can express the Taylor series of f_t centered at S_0 as

$$f_t(S_0) + m(S_t - S_0) + n(S_t - S_0)^2,$$

where

$$m = f'_t(S_0) \quad \text{and} \quad n = \frac{f''_t(S_0)}{2}.$$

Hedging

Let's hedge out the first order equity exposure by shorting m units of the underlying equity. The total gain/loss (ignoring fees) of the hedged position is

$$\begin{aligned} f_t(S_t) - f_t(S_0) - m(S_t - S_0) &\approx m(S_t - S_0) + n(S_t - S_0)^2 - m(S_t - S_0) \\ &= n(S_t - S_0)^2. \end{aligned}$$

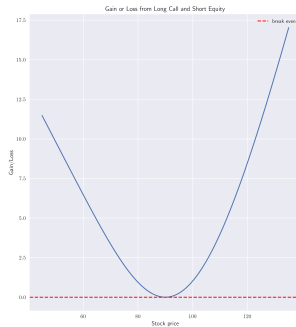
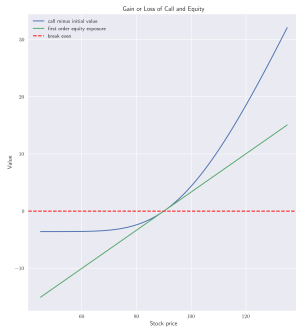
It Looks Like We Can't Lose!

Since $n = f_t''(S_0)/2 \geq 0$, the gain or loss

$$f_t(S_t) - f_t(S_0) - m(S_t - S_0) \approx n(S_t - S_0)^2 \geq 0.$$

It looks like we can't lose! There is a third order equity relationship, but it's not big enough to obtain a loss.

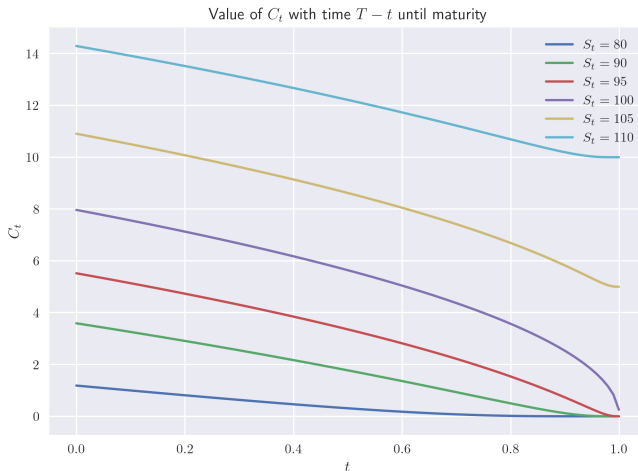
It Looks Like We Can't Lose!



No Such Thing as a Free Lunch

This analysis fails to consider the effect of time. As time passes, the value of C_t decays assuming the stock price S_t remains constant.

No Such Thing as a Free Lunch



Time Decay

We looked at the loss or gain of the position at time $t = 0$. However, as we saw in the previous slide, the value of the call decays over time. So, if $S_t \approx S_0$ and $t > 0$, our position loses money. Let's re-examine the gain/loss profile at a later time, specifically at $t = 0.6$.

Time Decay

